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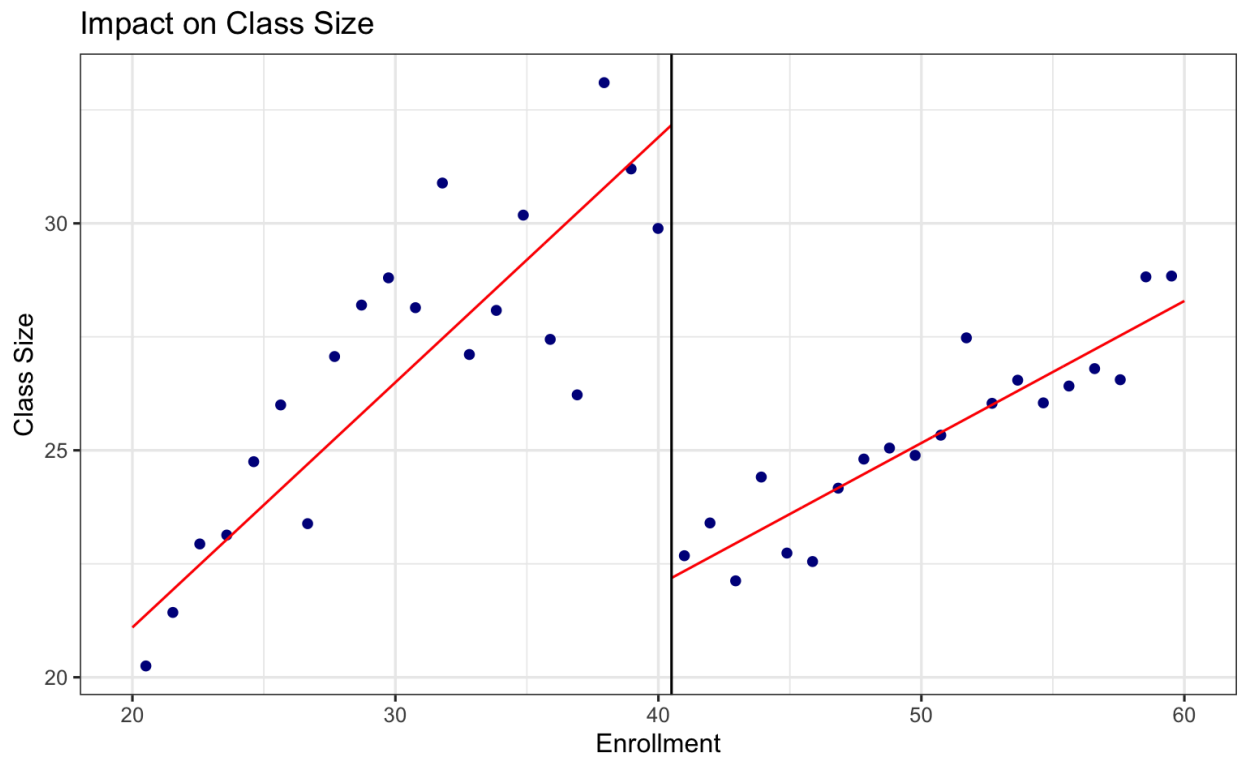
Class Size Policy Analysis for Conejo Valley Unified School District

I have compiled an assortment of empirics and information regarding class sizes, student enrollment, and student performance to help make an informed decision on class size reduction policies for this school district.

Firstly, I would like to acknowledge some constraints and establish why the data must be scrutinized very meticulously throughout this analysis. Simply comparing the test scores of students in small classes against the test scores of students in large classes will not illustrate the full picture of this dynamic: there are other factors at play that are not captured within this simple model. These omitted variables can potentially (and falsely) create a deceiving relationship between class size and student performance alone. For example, smaller private schools with inherently smaller class sizes may have greater funding for educators, differing curriculum, etc. Additionally, some areas may have more children who come from fortunate backgrounds, who are already predisposed to being successful academically and are less sensitive to factors such as teacher quality. Schools cannot be blindly compared to each other.

Upon observing class sizes with relation to total school enrollment for schools with a class size policy of 40 students maximum, some interesting conclusions can be drawn. The following figure is a binned scatterplot, allowing us to see data points within multiple "bins", one visualizing data before the 40 student cutoff, and one visualizing data beyond

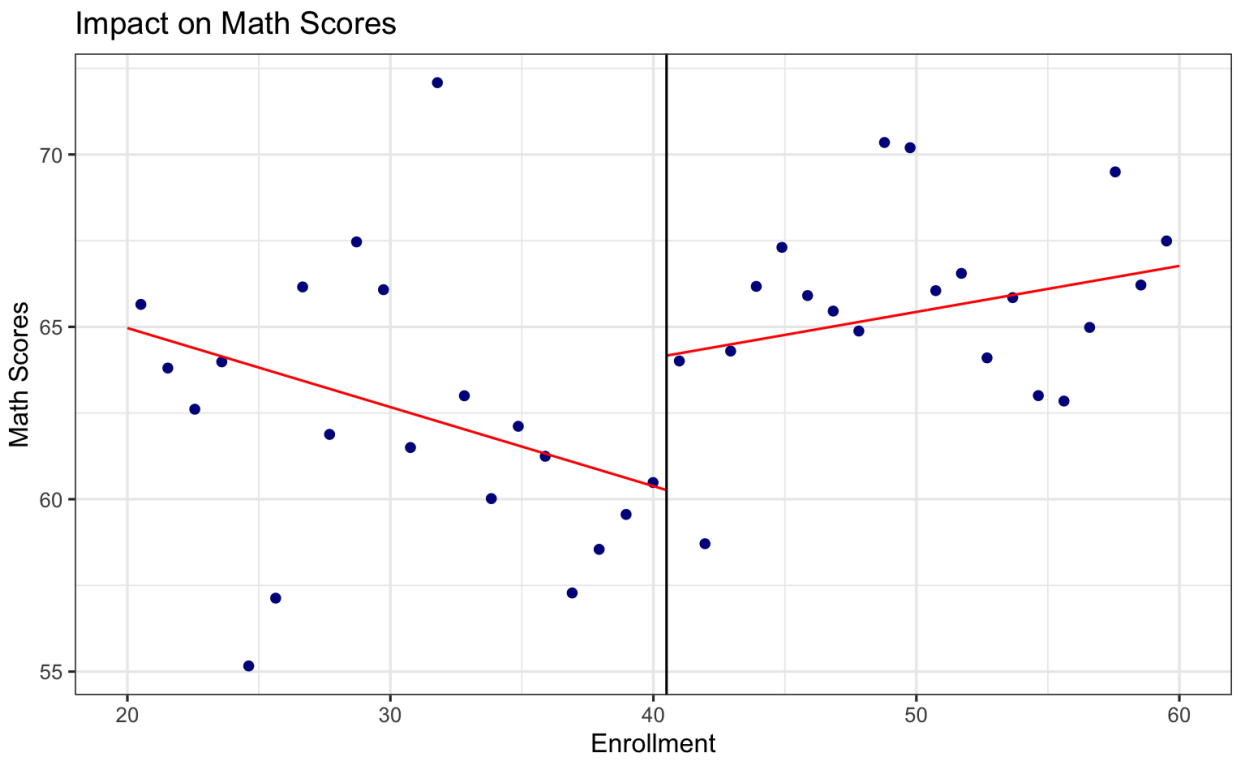
the 40 student cutoff.



We can see that, up to the 40 student level, as enrollment in schools increases, class size consistently increases as well. However, when enrollment exceeds the 40 student class size level, class sizes drop drastically, implying an additional class opened up, and the students were split into the two classes to stay below the maximum class size level. The regression estimates a threshold coefficient of -11.00075. This means we estimate class sizes drop by 11.00075 when the enrollment exceeds the 40 student limit.

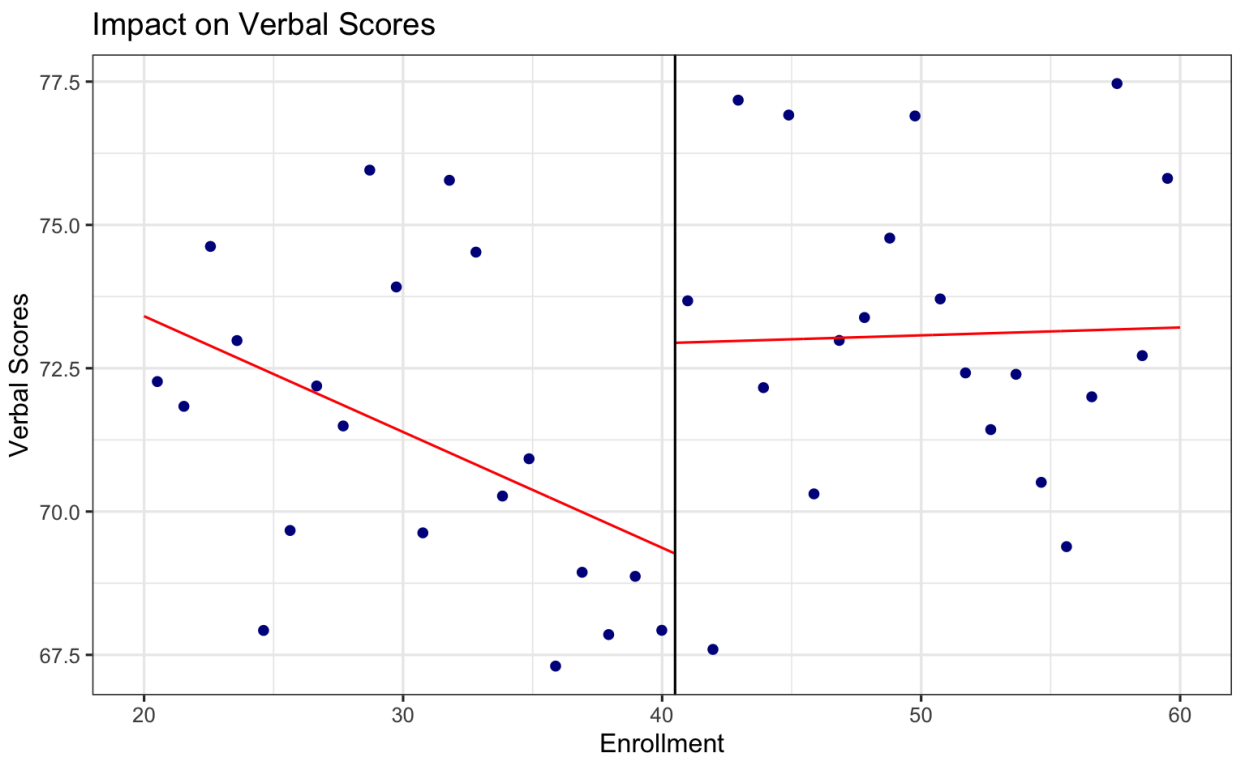
Other useful figures include models of enrollment with relation to various average test scores for students at the above observed schools. Here is a model of enrollment with

relation to average math test scores:



This regression estimates that average math scores drop by 0.19263 for each additional student in the class below the 40 student threshold. However, when the threshold is exceeded and class sizes split, the regression estimates that average math scores will increase by 3.4289. This does generate some association between smaller class sizes and increasing student performance.

An additional figure that looks similar to the one above visualizes a model of enrollment with relation to average verbal test scores:



This regression estimates a coefficient -0.16438 for average verbal scores of students in schools with enrollment below the 40 student threshold. This means that average verbal scores drop by 0.16438 for each additional student in the class below the 40 student threshold. However, when the threshold is exceeded and class sizes split, the regression estimates that average verbal scores will increase by 2.62813. While this association is not as strong as the association with math scores, this still does generate some association between smaller class sizes and increasing student performance.

If all schools followed the class size rule perfectly, we would expect class sizes to effectively cut in half as soon as they cross the 40 student enrollment boundary. For example, if a school has exactly 40 students enrolled, there would be one class at the school with 40 students. However (assuming they perfectly follow the rule), if one additional student enrolled at that school, they would need to open a second class. We would expect

20 students in one class and 21 students in the other. However, since we do not see the regression estimating a 20-21 class size drop when crossing the threshold, it may be possible that the schools are not all perfectly complying with the 40 student class limit policy.

The identification assumption for regression discontinuity in this case is that all schools just above and below the 40 student threshold are balanced and comparable in other parameters, meaning changes in student performance can only be explained by class size changes.

Enacting policies to reduce class sizes should still be handled cautiously. The data, along with external research, suggest that there may be different marginal returns from changes in different class sizes. This means that changes in larger classes could be more impactful on student learning than changes in small classes. As an example, a reduction from a class with 40 students down to 35 students might have significant impacts on student learning. However, a reduction from a class with 20 students down to 15 students may result in less pronounced impacts on student learning. While the reduction is 5 students in both cases, the efficacy of the policy is less with groups of students that are already small, as compared to larger classes that may feel a higher benefit from the exact same reduction.

Based on Jepsen and Rivkin's (2009) study of California's class size reduction program, there are potential risks that should be considered when deciding how, and if, class size reduction policies should be enacted in this district. The underlying problem with reducing class sizes is that it increases the district's demand for teachers. This, in turn, places a financial burden on the district to hire more teachers. Furthermore, it is possible

that increasing the amount of teachers at the district will actually lower the success of students. If the district hires lower quality educators due to their rapid need for more teachers, students will not find greater academic success, even in smaller classes.

Based on all of this information gathered, I would recommend that the CVUSD enacts a class size reduction policy. High school classes in this district fill up with upwards of 45 students. Reducing the amount of students in classes like this could show large benefits to those students. Furthermore, the district's budget is large and can be reallocated from expenditure on athletics and other departments that are less aligned with the district's goals: educating students. CVUSD has the financial capability to hire quality educators to take over the newly established classes, and the students have a need for more focused, smaller classes.